

Soham Gandhi

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EXPERIENCE

- Texas Instruments** | *Digital IC Design Engineer* | Santa Clara, CA Jul 2025 – Present
- Leading RTL development of a bidirectional control channel over SERDES interfaces.
 - Integrating mixed-signal Verilog models, bringing analog behavior into digital simulation environment.
 - Streamlining the development environment via a Git-based IP design and automated compilation.
 - Increasing assertion coverage and implemented formal connectivity verification, reducing design bugs by 25%.
 - Designing and implementing IP compliant with functional safety requirements, including systematic analysis and mitigation of all identified failure modes.
- Texas Instruments** | *Digital IC Design Intern* | Santa Clara, CA May 2024 – Aug 2024
- Executed RTL simulations and debugging for automotive SerDes chips improving audio/video functionality.
 - Designed a high-speed FIFO in SystemVerilog with prefetching logic and single-port RAM, improving performance by 50%.
- Texas Instruments** | *Validation Engineering Intern* | Santa Clara, CA May 2023 – Aug 2023
- Collaborated with an external company to develop a lab methodology that emulated automotive high-speed SerDes cables using a programmable Artek Variable ISI channel, cutting SerDes link testing time by 60%.
 - Automated ASIC setup and testbench configuration using Python, reducing validation time and effort by 3x.
- General Dynamics** | *Software Engineering Intern* | Fair Lakes, VA May 2022 – Aug 2022
- Automated Linux OS deployment over NFS using DHCP and TFTP, cutting deployment time by 50%.

EDUCATION

- Virginia Tech** Aug 2021 – May 2025
Dual B.S. in Computer & Electrical Engineering **GPA: 3.92**
Minor in Computer Science
- Relevant Coursework:** Adv Digital Design, Adv Computer Architecture, Design for Test, AI, Computer Vision
- Awards:** Calhoun Scholar, Dean's List, HackViolet 2023 (Best Innovation Hack)

PROJECTS

- FPGA Chess** | *Developer* Jan 2025 – May 2025
- Implemented a two-player FPGA-based chess system with move synchronization & countdown timers.
 - Designed and verified a parameterizable UART protocol enabling reliable, low-latency communication.
 - Constrained and synchronized three clock domains (VGA, Game Logic, UART), resolving CDC violations ensuring stable system performance.
- ARM Pointer Authentication Security** | *Research Assistant* Jun 2024 – May 2025
- Enhanced the Rocket Chip architecture, developed by UC Berkley, by integrating a custom pointer encryption algorithm that executes within a single CPU clock cycle.
 - Engineered hardware-level implementation of the PRINCE encryption algorithm.
 - Working towards publication to present advancements in pointer security and encryption techniques.
- BEARHW (sPACtre)** | *Research Assistant* Aug 2023 – May 2024
- Analyzed computer specifications and replicated 4 Spectre variant side-channel attacks on M1 Mac ARM architecture using C and Assembly, contributing to pending conference publication.
 - Investigated Spectre side-channel vulnerabilities and proposed mitigation strategies.
- Latis** | *Founder & Developer* Nov 2022 – May 2023
- Built a decentralized update platform using blockchain, reducing vulnerability to DDoS attacks by 70%.
 - Developed a web application to streamline manufacturer-OEM interactions via smart contracts, improving update times by 50%.

SKILLS

Programming Languages: SystemVerilog, C++, C, Python, Java, Bash, Git

Hardware Design: FPGA Design, ASIC Design, Clock Domain Crossing (CDC), SERDES, Mixed-Signal Design

Tools & Technologies: Cadence, Formal Verification, UVM, Intel Quartus, ModelSim, Keysight PNA